

QUICKBORN, Germany

WMO: 101460

Lat: 53.7331N

Lon: 9.8775E

Elev: 40

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 96-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.5	18.7	9.6	9.0	18.0	14.5	11.5	21.9	24.5	45.7	21.8	44.5	3.7	80	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	82.9	66.0	79.2	65.2	75.8	63.8	68.7	78.0	66.9	75.4	65.3	72.9	7.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	94.6	71.3	63.9	89.0	69.6	62.3	84.2	68.1	32.9	78.4	31.4	75.5	30.2	73.0	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.7	16.8	14.8	DB	7.6	89.8	8.4	3.3	1.6	92.2	-3.3	94.2	-8.1	96.1	-14.2	98.5
			WB	7.3	72.3	8.3	1.3	1.3	73.2	-3.6	74.0	-8.3	74.7	-14.3	75.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	34.8	35.7	39.5	47.1	53.8	58.7	63.0	62.6	56.8	49.3	41.3	35.1	(6)
	DBStd	12.24	8.44	7.46	7.16	6.83	6.01	5.74	5.25	4.99	5.41	6.02	6.76	8.09	(7)
	HDD50	2188	472	401	329	135	29	2	0	0	6	86	266	462	(8)
	HDD65	6229	937	820	790	537	351	206	104	109	253	486	710	926	(9)
	CDD50	1535	0	1	3	48	147	263	402	391	208	65	6	1	(10)
	CDD65	101	0	0	0	0	3	16	42	34	6	0	0	0	(11)
	CDH74	1120	0	0	0	7	60	188	483	323	58	1	0	0	(12)
	CDH80	255	0	0	0	0	9	39	129	74	4	0	0	0	(13)
(14) Wind	WSAvg	6.7	7.4	8.0	7.9	7.3	6.5	6.3	5.9	5.6	5.4	6.4	6.5	7.0	(14)
(15) Precipitation	PrecAvg	31.4	2.6	2.2	2.2	1.6	2.4	3.0	3.4	3.1	2.7	2.5	2.7	3.0	(15)
	PrecMax	42.9	5.4	5.8	4.6	3.3	6.1	6.4	8.1	7.4	8.0	7.3	5.5	6.9	(16)
	PrecMin	20.5	0.0	0.4	0.3	0.1	0.4	0.4	1.1	1.2	1.0	1.1	0.3	0.9	(17)
	PrecStd	5.1	1.4	1.2	1.2	0.8	1.1	1.5	1.8	1.5	1.6	1.3	1.3	1.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	54.2	62.4	74.4	80.1	84.8	89.1	86.2	80.0	67.2	58.5	53.7	(19)
		MCWB	50.2	48.9	52.2	58.7	64.0	68.2	67.5	66.7	64.7	60.8	54.5	51.0	(20)
	2%	DB	50.4	51.0	57.2	68.6	74.5	79.4	83.5	81.6	75.0	63.2	55.3	50.5	(21)
		MCWB	47.8	47.9	49.4	55.3	60.7	64.9	66.4	66.0	63.5	58.0	52.4	48.5	(22)
	5%	DB	47.8	48.6	53.1	64.4	70.4	74.7	79.4	77.3	70.5	60.6	52.9	48.0	(23)
		MCWB	45.7	46.3	48.4	53.4	58.5	63.0	65.6	64.9	61.2	56.7	50.5	46.0	(24)
	10%	DB	45.7	46.1	50.0	59.8	66.4	70.7	75.4	73.5	66.7	58.5	50.6	45.9	(25)
		MCWB	43.9	44.0	45.9	51.6	56.9	60.9	64.0	63.8	59.7	55.3	48.4	44.4	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.9	50.8	53.9	60.5	66.3	70.2	71.0	70.9	67.1	61.9	56.5	52.3	(27)
		MCDB	52.6	52.6	59.0	69.9	76.0	80.5	81.2	80.6	75.4	66.3	57.7	53.8	(28)
	2%	WB	48.4	48.6	51.3	56.9	63.0	67.1	68.8	68.1	64.5	59.2	53.0	48.7	(29)
		MCDB	50.2	50.2	54.8	65.7	71.3	76.2	78.7	76.3	71.8	62.0	54.9	50.2	(30)
	5%	WB	45.9	46.5	48.9	54.6	60.2	64.6	66.9	66.3	62.5	57.4	50.7	46.4	(31)
		MCDB	47.3	48.5	52.7	62.2	67.1	71.7	76.3	73.5	68.5	59.9	52.5	47.6	(32)
	10%	WB	44.0	44.0	46.4	52.4	57.8	62.3	65.2	64.7	60.8	55.6	48.6	44.4	(33)
		MCDB	45.5	45.9	49.4	58.9	64.3	68.6	73.2	71.5	65.4	58.1	50.3	45.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.3	9.5	13.4	18.1	19.2	18.4	18.4	17.9	16.8	12.9	9.2	8.1	(35)
		MCDBR	8.9	10.7	19.6	26.0	26.9	25.8	26.9	24.7	22.9	14.8	9.9	8.6	(36)
	5% WB	MCWBR	7.9	8.3	13.3	14.1	14.8	13.4	12.2	11.6	12.7	10.4	8.0	7.6	(37)
		MCWBR	8.5	9.6	16.7	22.0	22.7	22.5	23.5	20.3	19.7	13.5	9.4	8.6	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.318	0.337	0.367	0.419	0.411	0.409	0.428	0.421	0.393	0.355	0.333	0.310	(40)
		taud	2.340	2.345	2.291	2.178	2.243	2.276	2.240	2.271	2.333	2.426	2.386	2.334	(41)
	Ebn at Noon	Ebn at Noon	204	239	256	256	265	267	259	253	245	230	196	182	(42)
		Edh at Noon	18	25	32	41	41	40	41	37	31	23	18	16	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	176	388	771	1279	1608	1684	1595	1316	954	524	221	132	(44)
	RadStd	RadStd	21	54	76	159	182	132	211	98	108	60	23	9	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-257	N/A	N/A

Nomenclature: See separate page